

Clustering Adolescent Scooter Crashes in Île-de-France

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Research question

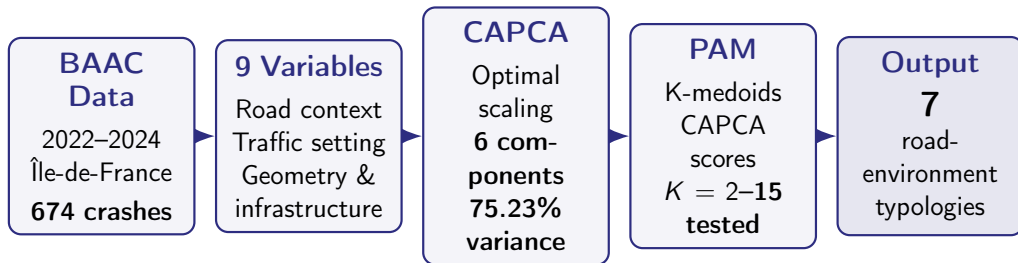
Can adolescent scooter crashes be grouped into
distinct **road-environment contexts**?

BAAC 2022–2024

Île-de-France • Ages 10–19 • 674 crashes



From Crash Data to Road-Environment Typologies



Injury severity was excluded from clustering to obtain severity-independent road-environment typologies.

Seven Road-Environment Crash Typologies

Seven Road-Environment Crash Typologies

Cluster	Typology	n (%)	Main distinguishing characteristics
C1	Departmental two-way roadway corridors	128 (19.0%)	Departmental road · Two-way traffic · Flat · Straight
C2	Municipal one-way intersection environments	102 (15.1%)	One-way traffic · Municipal road · Intersection · On roadway
C3	Municipal two-way intersection environments	120 (17.8%)	Intersection · Municipal road · Two-way traffic
C4	Municipal non-intersection roadway environments	126 (18.7%)	Outside intersection · On roadway · Flat · Straight
C5	Non-flat roadway environments	72 (10.7%)	Non-flat profile · Outside intersection · Cycle path
C6	Pavement and off-road environments	77 (11.4%)	Pavement crash position · Other reserved lane · Pedestrian zone
C7	Non-straight complex intersection environments	49 (7.3%)	Non-straight alignment · Designed intersection · Roundabout

The seven clusters represent distinct combinations of road type, intersection context, traffic configuration, geometry and infrastructure.

Distinct road-environment profiles emerge from adolescent scooter crashes, providing a basis for future injury-severity modelling with XGBoost and SHAP.

Visit the poster to discuss and exchange more about results and detailed cluster profiles.